

The Selfish/Altruistic Threshold in the Red/Blue Button Problem under Uncertainty

Matthew Sawoski

May 2026

Abstract

The Red/Blue button thought experiment recently went viral, prompting strongly polarized reactions. We analyze it through the pivotal-voter framework and derive a closed-form expression for the indifference threshold $u_n^*(\bar{v}, \tau)$, defined as the ratio of self-utility to per-other-utility at which an agent is indifferent between voting red and voting blue. The threshold has three asymptotic regimes: $\sqrt{n}$ growth at the symmetric point with known voter beliefs (recovering the bulk-regime calculation circulating informally), an n -independent asymptote $u_\infty^*(\bar{v}, \tau) = (2\tau)^{-1}\phi(\zeta)/\Phi(-\zeta)$ when the population’s blue-leaning rate is itself uncertain (where $\zeta = (\bar{v} - 1/2)/\tau$ is the standardized distance from the prior mean to the knife edge, and ϕ, Φ are the standard normal PDF and CDF), and a sharp crossover between them at $n \sim 1/(4\tau^2)$. The closed-form u_∞^* specializes a result of Edlin–Gelman–Kaplan to the Red/Blue puzzle and exhibits the dominant role played by uncertainty τ over the central estimate $\bar{v}$. We also analyze a small- n regime in which particular attachments to specific individuals can flip the decision. The unified formula u_n^* is suitable as a ground-truth for empirical experiments: given n , $\bar{v}$, and τ , a respondent’s choice between red and blue reveals whether their w_0/w ratio exceeds the threshold.

1 Introduction

On April 24, 2026, the writer Tim Urban posted the following thought experiment to X (Urban, 2026):

Everyone in the world has to take a private vote by pressing a red or blue button. If more than 50% of people press the blue button, everyone survives. If less than 50% of people press the blue button, only people who pressed the red button survive. Which button would you press?

The post drew over 22 million views within three days. The accompanying poll closed with 57.9% blue and 42.1% red. Mainstream and technical commentary followed: *The Conversation*, *Reason*, *Medium*, and a wave of Substack and *Hacker News* discussions appeared in the subsequent week, with respondents on both sides claiming their answer was the obviously correct one and the opposite was self-evidently wrong.

The standard “vote red” argument is the dominance argument: voting red guarantees personal survival regardless of the aggregate outcome, while voting blue saves you only conditional on a coalition of fellow blue voters that you cannot influence. The standard “vote blue” argument is altruistic and coalitional: if a sufficient fraction of the population reasons the same way, everyone lives, and the cost of guarding against defection by always voting red is the loss of any cooperative equilibrium worth having.

Among the technical writeups that have appeared since the puzzle went viral, Graber (2026) performs the symmetric-coin-flip calculation and shows that at $v = 1/2$ the expected number of lives saved by voting blue scales as $\sqrt{n/\pi}$, but is fragile to deviations from $v = 1/2$. Krue (2026) introduces an error-rate framework in which mild family-directed altruism can rationalize blue around an all-blue equilibrium. Chappell (2026) compares the puzzle to variants such as Roko’s Blender and concludes that the answer depends on framing-sensitive empirical predictions about others’ behavior.

The academic precursor in the rational-choice voting literature is Edlin et al. (2007), who derive that the expected utility of voting in a large election does not vanish with n once a voter has socially-directed preferences and the perceived closeness of the election does not depend strongly on the population size. The key mechanism, that the pivotal probability scales as $1/n$ rather than $1/\sqrt{n}$ once one has nontrivial uncertainty about the underlying voting fraction, is precisely the structure that produces our n -independent epistemic asymptote. Related work includes Gelman et al. (2009) on the empirical frequency of pivotal votes in U.S. elections, and a long literature on calculus-of-voting models going back to Riker and Ordeshook (1968).

Our contribution is a complete and explicit phase diagram for $u_n^*(\bar{v}, \tau)$: a single closed-form expression that handles both bulk and epistemic regimes uniformly, exhibits the correct asymptotic structure in each, and reduces to known results in the appropriate limits. The contribution relative to Graber (2026) is the inclusion of epistemic uncertainty and the bulk-to-epistemic crossover; relative to Edlin et al. (2007) is the explicit small- n behavior and the application to Red/Blue specifically; relative to Krue (2026) is a treatment of uncertainty over v rather than per-vote error rates. We also derive the population scales at which particular attachments to specific individuals can flip the decision, a natural extension that makes the analysis applicable to small-group variants of the puzzle.

2 Setup and the Pivotal-Voter Framework

Let the population consist of $n+1$ persons $p_0, p_1, \dots, p_n$. We are person p_0 and must decide whether to press red or blue. For each person p_i , we assign a utility $w_i \geq 0$ representing how we value their survival; in particular w_0 is our own self-utility. For each $i \neq 0$, we assign a probability $v_i \in [0, 1]$ representing our subjective belief that p_i presses blue. Throughout, we take n to be even so that the total population is odd and exact ties are not possible; the survival rule “strict majority blue” then resolves cleanly.

Let $X = \sum_{i=1}^n B_i$ be the number of blue votes among the others, where $B_i = \mathbf{1}\{p_i \text{ votes blue}\}$. The event “strict majority blue” becomes $X + \mathbf{1}\{p_0 \text{ blue}\} > (n+1)/2$, which simplifies to:

- If you vote blue: everyone survives iff $X \geq n/2$.
- If you vote red: everyone survives iff $X \geq n/2 + 1$.

Thus $X = n/2$ is the unique *pivotal* event, where your vote alone determines whether everyone lives.

Decomposing the difference in expected utility outcome by outcome:

$X > n/2$: Blue wins regardless. No utility difference.

$X < n/2$: Red wins regardless. The blue voters among the others die in both cases, so their utilities

cancel. The only difference is your fate: you live under red, die under blue. Conditional difference: $-w_0$.

$X = n/2$: Voting blue saves everyone (utility $\sum_{i=0}^n w_i$); voting red kills the blue voters among the others (utility $w_0 + \sum_{i:p_i \text{ red}} w_i$). Conditional difference: $\sum_{i:p_i \text{ blue}, i \neq 0} w_i$.

Taking expectations and using linearity in the pivotal sum,

$$\Delta U := \mathbb{E}[U(\text{blue}) - U(\text{red})] = \mathbb{P}(X = n/2) \sum_{i=1}^n w_i \mathbb{P}(p_i \text{ blue} \mid X = n/2) - \mathbb{P}(X < n/2) w_0. \quad (1)$$

Idealized symmetric case. We henceforth specialize to the case where all others are exchangeable: $w_i = w$ and $v_i = v$ for all $i \neq 0$. Exchangeability implies $\mathbb{P}(p_i \text{ blue} \mid X = n/2)$ is identical across i , and the values must sum to $\mathbb{E}[X \mid X = n/2] = n/2$. Therefore $\mathbb{P}(p_i \text{ blue} \mid X = n/2) = 1/2$, and the pivotal sum collapses to $w \cdot n/2$. Equation (1) becomes

$$\Delta U = \frac{n}{2} w \mathbb{P}(X = n/2) - \mathbb{P}(X < n/2) w_0. \quad (2)$$

Definition 1. The indifference threshold u_n^* is the ratio w_0/w at which $\Delta U = 0$. Explicitly,

$$u_n^*(\bar{v}, \tau) = \frac{n}{2} \cdot \frac{\mathbb{P}(X = n/2)}{\mathbb{P}(X < n/2)}. \quad (3)$$

A rational agent prefers blue to red iff $w_0/w < u_n^*$, where w_0 is the agent's self-valuation and w is the agent's per-stranger valuation. The remaining task is to evaluate the right side as a function of n , $\bar{v}$, and the prior uncertainty τ on v .

3 Main Result: The Indifference Threshold in the Symmetric Case

3.1 Bulk regime: known v

When v is known, $X \sim \text{Binomial}(n, v)$ exactly. For large n , the local central limit theorem gives

$$\mathbb{P}(X = m) \approx \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(m - \mu)^2}{2\sigma^2}\right), \quad \mathbb{P}(X \leq m) \approx \Phi\left(\frac{m + \frac{1}{2} - \mu}{\sigma}\right), \quad (4)$$

with $\mu = nv$ and $\sigma^2 = nv(1 - v)$. The continuity correction in the cumulative distribution is the standard $\pm \frac{1}{2}$; for the point probability, integrating the Gaussian density over a unit-width interval centered at m gives the density at the midpoint to leading order, with no half-integer shift.

Substituting into (3) and writing $z = (n/2 - \mu)/\sigma$,

$$u_n^* \approx \frac{n}{2} \cdot \frac{\phi(z)/\sigma}{\Phi(z - 1/(2\sigma))}. \quad (5)$$

Symmetric point. At $v = 1/2$ we have $z = 0$ and $\sigma = \sqrt{n}/2$, giving

$$u_n^* = \sqrt{\frac{2n}{\pi}} \cdot [1 + \mathcal{O}(n^{-1/2})]. \quad (6)$$

This is the result derived informally by Graber (2026). For $n = 8 \times 10^9$ (world population), $u_n^* \approx 71,366$.

Asymmetric points. For $v < 1/2$, $z \rightarrow +\infty$ as $n \rightarrow \infty$, and the Gaussian factor in the numerator decays as $e^{-z^2/2}$ while $\Phi(z) \rightarrow 1$. Hence u_n^* decays exponentially in n . For $v > 1/2$, $z \rightarrow -\infty$, and the Mills-ratio asymptotic $\Phi(z)/\phi(z) \sim 1/|z|$ as $z \rightarrow -\infty$ gives $u_n^* \sim n|v - 1/2|/(2v(1-v))$, linear in n . In both cases, the dichotomy reflects whether the tie is in the bulk of the binomial distribution or in its tails.

3.2 Epistemic regime: uncertain v

We now allow uncertainty about v via a hierarchical Bayesian model. Let $V \sim \pi$ with $\mathbb{E}V = \bar{v}$ and $\text{Var}V = \tau^2$, and assume the votes are conditionally independent and identically distributed Bernoulli(V) given V . The marginal moments of X are

$$\mathbb{E}X = n\bar{v}, \quad \text{Var}X = \mathbb{E}[\text{Var}(X | V)] + \text{Var}(\mathbb{E}[X | V]) = n\bar{v}(1 - \bar{v}) + n^2\tau^2 \cdot \frac{n-1}{n}. \quad (7)$$

Define $\sigma_{\text{tot}}^2 = n\bar{v}(1 - \bar{v}) + n(n-1)\tau^2$. The epistemic variance term scales as $n^2\tau^2$, and dominates the intrinsic sampling variance $\sim n$ once $n \gg 1/\tau^2$. The crossover scale is

$$n_{\times} \sim \frac{1}{4\tau^2}. \quad (8)$$

Below $n_{\times}$, the bulk-regime analysis applies. Above $n_{\times}$, the epistemic regime takes over.

Local CLT for the mixture. The local CLT continues to hold for the mixture X in the epistemic regime, with the substitution $\sigma \rightarrow \sigma_{\text{tot}}$. The cleanest derivation is via Laplace's method applied to

$$\mathbb{P}(X = m) = \int_0^1 \pi(v) \binom{n}{m} v^m (1-v)^{n-m} dv. \quad (9)$$

The integrand is a product of two functions: the prior $\pi(v)$ with width τ , and the binomial likelihood, which (viewed as a function of v for fixed m) is Gaussian-shaped around $v = m/n$ with width $\sim 1/\sqrt{n}$. For $\tau \gg 1/\sqrt{n}$ the prior is broader than the likelihood, and the integral concentrates around $v = m/n$, giving

$$\mathbb{P}(X = m) \approx \frac{\pi(m/n)}{n} \cdot [1 + \mathcal{O}(1/(n\tau^2))]. \quad (10)$$

In particular, $\mathbb{P}(X = n/2) \approx \pi(1/2)/n$, scaling as $1/n$ rather than $1/\sqrt{n}$. For the cumulative,

$$\mathbb{P}(X < n/2) \rightarrow \pi([0, 1/2)) \quad \text{as } n \rightarrow \infty. \quad (11)$$

The n -independent asymptote. Substituting into (3),

$$u_{\infty}^*(\bar{v}, \tau) = \lim_{n \rightarrow \infty} u_n^* = \frac{1}{2} \cdot \frac{\pi(1/2)}{\pi([0, 1/2))}. \quad (12)$$

This limit is independent of n . Specializing to a Gaussian prior $V \sim \mathcal{N}(\bar{v}, \tau^2)$ (truncated to $[0, 1]$, with τ small enough that the truncation is negligible), we have $\pi(1/2) = (1/(\tau\sqrt{2\pi})) \exp(-(\bar{v} - 1/2)^2/(2\tau^2))$ and $\pi([0, 1/2)) = \Phi(-\zeta)$, where

$$\zeta = \frac{\bar{v} - 1/2}{\tau} \quad (13)$$

is the standardized distance from the prior mean to the knife edge. After cancellation, the threshold takes the closed form

$$u_{\infty}^*(\bar{v}, \tau) = \frac{1}{2\tau} \cdot \frac{\phi(\zeta)}{\Phi(-\zeta)}, \quad (14)$$

where ϕ and Φ are the standard normal PDF and CDF, and $\phi(\zeta)/\Phi(-\zeta)$ is the standard normal hazard rate evaluated at $-\zeta$.

This is the central formula of the paper. The crucial dependence is on τ : the threshold scales as $1/\tau$ at the knife edge and as $1/\tau^2$ in the blue-leaning regime, with $\bar{v}$ entering only through the dimensionless parameter ζ . Uncertainty about the population's voting fraction is the dominant input, far more so than the central estimate itself once $|\bar{v} - 1/2|$ is small compared to τ .

Sub-regimes in ζ . The closed form (14) produces three qualitatively distinct cases:

Knife-edge ($\zeta = 0$). $\phi(0)/\Phi(0) = \sqrt{2/\pi}$, giving the symmetric formula $u_{\infty}^* = 1/(\tau\sqrt{2\pi}) \approx 0.399/\tau$. Concretely, $\tau = 0.01$ gives $u_{\infty}^* \approx 40$, $\tau = 0.05$ gives $u_{\infty}^* \approx 8$, and $\tau = 0.1$ gives $u_{\infty}^* \approx 4$.

Population believed to lean red ($\zeta \ll -1$). The numerator $\phi(\zeta)$ decays as $e^{-\zeta^2/2}$ while $\Phi(-\zeta) \rightarrow 1$, so $u_{\infty}^* \approx \phi(\zeta)/(2\tau)$, exponentially small in ζ^2 . Almost any agent prefers red, decisively recovering the dominance conclusion. There is no plausible utility function that votes blue here.

Population believed to lean blue ($\zeta \gg 1$). The Mills-ratio asymptotic $\phi(\zeta)/\Phi(-\zeta) \sim \zeta + 1/\zeta + \dots$ gives

$$u_{\infty}^* \sim \frac{\zeta}{2\tau} = \frac{\bar{v} - 1/2}{2\tau^2}. \quad (15)$$

The threshold grows linearly in $\bar{v} - 1/2$ and inversely in τ^2 . Most agents vote blue, since under confident blue-leaning beliefs voting blue is nearly free: you almost certainly survive as part of the majority, and the small additional pivotal upside is a bonus.

The structure mirrors the bulk regime's three-way dichotomy, with $1/\tau$ playing the role of $\sqrt{n}$. In the bulk regime, larger populations push agents toward blue at the knife edge but extreme red below it and extreme blue above it; in the epistemic regime, smaller τ does the analogous work. Uncertainty acts as the inverse of an effective population scale.

Numerical values: knife edge versus the X poll. The X poll closed with 57.9% of $\sim 95,000$ respondents choosing blue. Taking this as a (caveated) estimate of $\bar{v}$ gives $\bar{v} = 0.579$, while the symmetric knife-edge case gives $\bar{v} = 0.500$. Table 1 reports u_{∞}^* for each combination of τ and $\bar{v}$, with each row labeled by the agent's epistemic state in plain language. The caveats on $\bar{v} = 0.579$ are substantial (the poll is self-selected, restricted to English-speaking X users, asks about a hypothetical scenario that may not predict actual behavior, and is subject to the framing-as-coordination-device effect discussed in §5); we include it because it illustrates how strongly the threshold depends on whether $\bar{v}$ sits at or even slightly above the knife edge.

Epistemic state	“I’m 95% sure v is in...”	u_∞^*
$\tau = 0.01, \bar{v} = 0.500$	[0.48, 0.52]	40
$\tau = 0.01, \bar{v} = 0.579$	[0.56, 0.60]	401
$\tau = 0.05, \bar{v} = 0.500$	[0.40, 0.60]	8
$\tau = 0.05, \bar{v} = 0.579$	[0.48, 0.68]	20
$\tau = 0.10, \bar{v} = 0.500$	[0.30, 0.70]	4
$\tau = 0.10, \bar{v} = 0.579$	[0.38, 0.77]	7
$\tau = 0.20, \bar{v} = 0.500$	[0.11, 0.89]	2
$\tau = 0.20, \bar{v} = 0.579$	[0.19, 0.97]	3

Table 1: Numerical values of the epistemic asymptotic threshold $u_\infty^*(\bar{v}, \tau)$ from (14). Each row is one possible epistemic state, with the prior 95% credible interval $\bar{v} \pm 1.96\tau$ written out for interpretability. An agent should vote blue if their self-valuation in units of stranger-utility falls below u_∞^* , red if above. Within each τ block, comparing the two rows shows how a 0.079 shift in the central estimate of $\bar{v}$ changes the threshold: the shift is dramatic when τ is small (tight prior) and modest when τ is large.

3.3 Unified formula

A single Gaussian approximation handles both regimes when we substitute σ_{tot} for σ in (5):

$$u_n^*(\bar{v}, \tau) \approx \frac{n}{2} \cdot \frac{\phi(z)/\sigma_{\text{tot}}}{\Phi(z - 1/(2\sigma_{\text{tot}}))}, \quad z = \frac{n/2 - n\bar{v}}{\sigma_{\text{tot}}}, \quad \sigma_{\text{tot}}^2 = n\bar{v}(1 - \bar{v}) + n(n - 1)\tau^2. \quad (16)$$

This expression reduces to (5) when $\tau = 0$, recovers (12) as $n \rightarrow \infty$ for $\tau > 0$, exhibits the crossover (8), and is straightforward to evaluate numerically.

Rate of convergence to u_∞^* . For practical use it is useful to know how fast the finite- n threshold approaches its limit as a function of n relative to the crossover scale $n_\times = 1/(4\tau^2)$. At $\bar{v} = 1/2$, expanding (16) gives

$$\frac{u_n^*}{u_\infty^*} \approx \frac{1}{\sqrt{1 + n_\times/n}} \cdot [1 + \mathcal{O}(1/\sigma_{\text{tot}})], \quad (17)$$

where the leading factor comes from the ratio $n\tau/\sigma_{\text{tot}}$ and the correction from the Mills-ratio expansion of $\Phi(-1/(2\sigma_{\text{tot}}))$. Inverting:

$$n \geq 4.3 n_\times \Rightarrow u_n^* \geq 0.90 u_\infty^*, \quad n \geq 9.3 n_\times \Rightarrow 0.95 u_\infty^*, \quad n \geq 49 n_\times \Rightarrow 0.99 u_\infty^*.$$

In words: when the population is several times the crossover scale, the asymptotic threshold u_∞^* is already a good approximation. For $\tau = 0.05$ ($n_\times = 100$), the asymptotic regime is essentially complete by $n \approx 5,000$. For $\tau = 0.01$ ($n_\times = 2,500$), it is complete by $n \approx 10^5$. World-scale populations ($n \sim 10^{10}$) are in the asymptotic regime for any $\tau \geq 10^{-4}$, so the constant u_∞^* is the operationally relevant threshold for almost any practical analysis of the original puzzle. Conversely, for small groups ($n \lesssim 4n_\times$) the bulk-regime $\sqrt{n}$ growth dominates and one should use (16) or the exact binomial directly.

3.4 Phase diagram

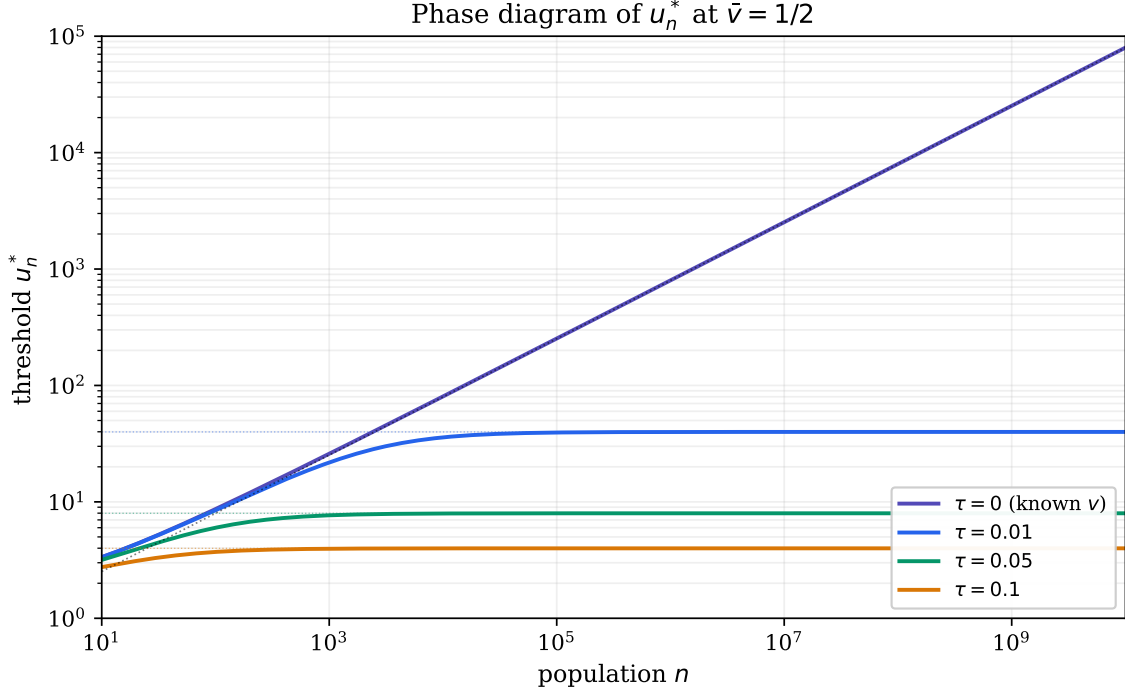


Figure 1: Indifference threshold u_n^* as a function of n at $\bar{v} = 1/2$, computed from the unified formula (16). The $\tau = 0$ curve grows as $\sqrt{2n/\pi}$. For $\tau > 0$, each curve initially follows the $\tau = 0$ trajectory, departs at $n \sim 1/(4\tau^2)$, and asymptotes to $u_\infty^* = 1/(\tau\sqrt{2\pi})$, indicated by the dotted horizontal lines. A rational agent votes blue if their self-valuation w_0/w falls below the curve, red if above.

Figure 1 shows u_n^* across nine orders of magnitude in n for several values of τ . The qualitative content of the phase diagram is summarized by Table 2.

Regime	Condition	Behavior of u_n^*
Bulk, symmetric	$\tau = 0, \bar{v} = 1/2$	$\sqrt{2n/\pi}$
Bulk, $\bar{v} < 1/2$	$\tau = 0, \bar{v} < 1/2$	exponentially decaying in n
Bulk, $\bar{v} > 1/2$	$\tau = 0, \bar{v} > 1/2$	linearly growing in n
<i>Epistemic regime</i> ($n \gg 1/\tau^2$): $u_\infty^*(\bar{v}, \tau) = (2\tau)^{-1} \phi(\zeta)/\Phi(-\zeta), \quad \zeta = (\bar{v} - 1/2)/\tau$		
knife-edge	$\zeta = 0$	$1/(\tau\sqrt{2\pi}) \approx 0.399/\tau$
leans red	$\zeta \ll -1$	$\approx \phi(\zeta)/(2\tau)$, exponentially small in ζ^2
leans blue	$\zeta \gg 1$	$\sim (\bar{v} - 1/2)/(2\tau^2)$, linear in $\bar{v} - 1/2$

Table 2: Asymptotic regimes of u_n^* . The standardized parameter $\zeta = (\bar{v} - 1/2)/\tau$ measures the prior mean's distance from the knife edge in units of prior standard deviation. The epistemic regime mirrors the bulk regime's three-way dichotomy with $1/\tau$ in the role of $\sqrt{n}$.

Interpretation. Two things are striking. First, the $\sqrt{n}$ growth at the symmetric known- v point is responsible for the apparent “vote blue dominates” conclusion at world scale: at $n \approx 8 \times 10^9$, the threshold is $u_n^* \approx 71,000$, far above any plausible self-valuation in units of stranger-utility. Second, this conclusion is fragile: any non-trivial uncertainty about v collapses the threshold to a small constant. A standard deviation of $\tau = 0.05$ on one’s belief about the population’s blue-leaning rate produces $u_\infty^* \approx 8$, and the transition from $\sqrt{n}$ growth to n -independent asymptote already takes place at $n_\times \approx 100$. The dramatic “world-scale altruism dominates self-interest” conclusion lives on a measure-zero set in the parameter space of priors over v .

This sensitivity is the substantive point of the analysis. The standard “vote red” dominance argument is correct under any prior with non-negligible uncertainty about the population’s mood. The “vote blue” rebuttal recovers force only under the assumption that one knows, with high confidence, that the population’s blue-leaning rate is exactly at or above the knife edge. Realistic epistemic states put even reasonably altruistic agents back into the red regime.

4 Particular Attachments

The idealized symmetric case treats all others identically. We now relax this by introducing a small set S of “special” people (such as immediate family) for whom the self has distinct utilities w_{s_j} and beliefs v_{s_j} . The total population becomes $n + |S| + 1$, with $|S| \ll n$ so that the bulk of the analysis above carries over.

4.1 General formula with attachments

Let X_n and X_S be the blue-vote counts among the n background and $|S|$ special voters respectively, so $X = X_n + X_S$. Pivotal at $X = N/2$ where $N = n + |S|$. The conditional probabilities split:

- Background voters retain $\mathbb{P}(p_i \text{ blue} \mid X = N/2) \rightarrow \frac{1}{2}$ in the epistemic regime and the symmetric bulk regime, by the exchangeability argument applied to the background subset.
- Special voters have $\mathbb{P}(p_{s_j} \text{ blue} \mid X = N/2) \approx v_{s_j}$, since their own probability V_{s_j} is independent of the population parameter V and conditioning on the tie is uninformative about V_{s_j} to leading order in $|S|/\sqrt{n}$ or $|S|/n$.

This leads to a clean additive correction to the pivotal sum:

$$\mathbb{E} \left[\sum_{i: \text{blue}, i \neq 0} w_i \mid X = N/2 \right] \approx w \cdot \frac{N}{2} + \sum_{j=1}^{|S|} (w_{s_j} - w) v_{s_j}. \quad (18)$$

The first term is what we would obtain under full homogeneity, and the second is an additive correction representing the *excess* utility each special person brings beyond the background, weighted by their marginal probability of voting blue. The threshold becomes

$$u_n^* \approx \frac{\mathbb{P}(X = N/2)}{\mathbb{P}(X < N/2)} \cdot \left[w \cdot \frac{N}{2} + \sum_j (w_{s_j} - w) v_{s_j} \right]. \quad (19)$$

An observation about τ_{s_j} . The special-person uncertainties τ_{s_j} do not appear in the leading-order threshold. The reason is structural: the contribution of p_{s_j} to ΔU is $w_{s_j} \mathbf{1}\{B_{s_j} = 1\}$ on the pivotal event, which is linear in the vote indicator. Linearity of expectation produces $w_{s_j} v_{s_j}$, with no role for higher moments of V_{s_j} . This would change if utility were non-linear in the survival of multiple special people (such as a hard preference for “all family lives”) or if special-person votes were correlated with V .

4.2 When special people change the answer at large n

Comparing the second term in the bracket to the first, the special-people correction matters when $\sum_j (w_{s_j} - w) v_{s_j} \gtrsim n \cdot w$. For world-scale $n \approx 8 \times 10^9$ and any handful of family members with finite valuations w_{s_j} , this is far from satisfied. Concretely, a spouse and three children each valued at 100 times an average stranger and each voting blue with probability $1/2$ contribute $\sum (w_{s_j} - w) v_{s_j} \approx 198$, and the relative shift in u_n^* is $\sim 10^{-7}$. Imperceptible.

4.3 When special people change the answer at small n

The story differs at small n , where pivotality is no longer a measure-zero event and a single concentrated stake can outweigh the gamble against personal survival. To make this concrete, consider the limit $w = 0$ (one cares only about self and special people) with a single special person, the spouse, valued at w_s and known to vote blue with $v_s = 1$. The threshold becomes

$$u_n^{\text{spouse*}} \approx \frac{\mathbb{P}(X_n < (n-1)/2)}{\mathbb{P}(X_n = (n-1)/2)}, \quad (20)$$

where X_n is the blue count among the n background voters.

Asymptotics. In the bulk regime at $\bar{v} = 1/2$, the binomial gives

$$u_n^{\text{spouse*}} \sim \sqrt{\frac{\pi n}{8}}. \quad (21)$$

In the epistemic regime, the pivotal probability scales as $1/n$ instead of $1/\sqrt{n}$, while the red-wins probability remains $\sim \frac{1}{2}$, giving

$$u_n^{\text{spouse*}} \sim n\tau\sqrt{\pi/2}. \quad (22)$$

Both grow with n , so the spouse stops mattering at large enough population. The bulk and epistemic asymptotes cross at $n \sim 1/(8\tau^2)$, with the epistemic curve exceeding the bulk curve above this scale: under uncertainty, particular attachments lose their decision-flipping power at *smaller* populations than they would under known $v = 1/2$, because pivotal probability decays faster.

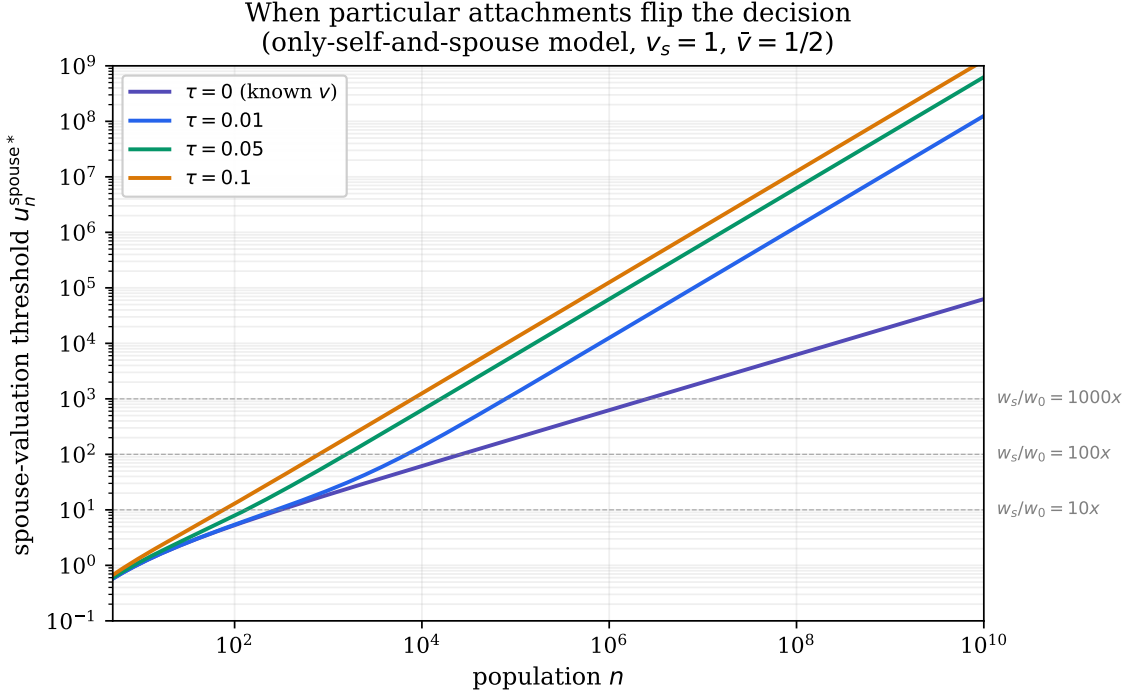


Figure 2: Spouse-valuation threshold $u_n^{\text{spouse*}}$ as a function of n , for several values of τ , in the only-self-and-spouse model with $v_s = 1$ and $\bar{v} = 1/2$ for the background. The spouse alone flips the decision to blue when w_s/w_0 exceeds the curve; below, the dominance argument prevails. Horizontal dashed lines mark typical valuations of $w_s/w_0 = 10, 100, 1000$.

Numerical thresholds. Figure 2 reads off the population at which the spouse ceases to matter, as a function of how strongly one values them. For $w_s/w_0 = 10$, the spouse flips the decision up to $n \approx 250$ in the bulk regime and $n \approx 160$ for $\tau = 0.05$. For $w_s/w_0 = 100$, the corresponding scales are $n \approx 2.5 \times 10^4$ and $n \approx 1.5 \times 10^3$. For $w_s/w_0 = 1000$, they are $n \approx 2.5 \times 10^6$ and $n \approx 1.5 \times 10^4$. At world scale, no realistic valuation works: the threshold $u_n^{\text{spouse*}}$ exceeds 10^9 in all the regimes shown.

The substantive conclusion: at twenty-person scale, the puzzle is fundamentally different from the world-scale version. A small group facing the buttons together has pivotal probability close to 1, and even mild attachments to specific group members can rationalize voting blue. At population scale, anonymity dominates: pivotal probability is too small for any concentrated personal stake to outweigh the guaranteed cost of being on the losing side, and the dominance argument controls.

5 Discussion and Empirical Use

The phase diagram organizes the existing arguments in the popular discourse and clarifies which is correct under which conditions.

The “vote red” dominance argument is correct in the regime that matches realistic epistemic states for a world-scale puzzle: when the population is large and the agent has even modest uncertainty about the underlying voting fraction. In this regime u_n^* is a small constant on the order of $1/\tau$, and a self-valuation greater than this constant in units of stranger-utility flips the agent to red.

The “vote blue” altruistic argument is correct under two distinct conditions. First, in small populations, where pivotal probability is large enough that even modest altruism (broad or particular) makes blue the expected-utility maximizer. Second, in the limiting case where the agent has a sharp prior pinning the population’s voting fraction to exactly 1/2: under that exact belief, u_n^* grows as $\sqrt{n}$ and outpaces any reasonable self-valuation. This second condition is mathematically real but epistemically unrealistic.

The third argument circulating in the discourse, that one’s choice is correlated with others’ choices and so should be made as if for a coalition of agents reasoning identically, is a decision-theoretic reframing rather than a parameter regime within our model. Our analysis is conducted within the expected-utility framework with causal independence, that is, causal decision theory. An agent committed to evidential decision theory, updateless decision theory, or functional decision theory will reach different conclusions; these alternatives are well-discussed in Kruegel (2026) and the broader rationalist literature on Newcomb-like problems and are outside the scope of this writeup.

Use in empirical experiments. The unified formula (16) is suitable as a ground-truth threshold against which to interpret experimental data on red/blue choices. In an experiment where n , $\bar{v}$, and τ are known or controlled, an agent’s choice between red and blue reveals whether their effective w_0/w ratio exceeds u_n^* . Under reasonable parameter ranges suggested by Figure 1, u_n^* falls in the single- to low-double-digit range, putting the threshold near the boundary of plausible self-valuations and giving the experiment power to distinguish self-interested from altruistic preferences.

The X poll itself, in which 57.9% of 95,500 respondents chose blue, can be read against Table 1. A respondent who took $\bar{v} = 0.579$ (the poll’s own number) and held a moderate uncertainty $\tau = 0.05$ would face a threshold $u_\infty^* \approx 20$; tightening to $\tau = 0.025$ pushes it to ≈ 69 , loosening to $\tau = 0.10$ drops it to ≈ 7 . The same respondent at $\bar{v} = 0.500$ would see thresholds half to a quarter of these. The point is that the agent’s choice depends sharply on which row of the table they actually inhabit, with plausible answers spanning two orders of magnitude. A natural alternative interpretation, raised by both Graber (2026) and Chappell (2026), is that respondents are not estimating $\bar{v}$ at all: under the framing of a poll, observing a leading blue tally on the screen functions as a soft coordinating mechanism that effectively pushes τ toward zero, restoring the bulk-regime $\sqrt{n}$ analysis under which voting blue dominates at world scale. The 57.9% blue rate may say less about respondents’ baseline self-valuations than about whether the poll itself functioned as an equilibrium-selection device, replacing the no-coordination assumption that gives the original puzzle its bite.

Limitations. The hierarchical Bayes model with shared V is one of two natural ways to formalize “uncertainty about the population’s voting fraction”; the alternative, in which each voter has an independent draw of V_i from a per-person prior, eliminates the $n^2\tau^2$ epistemic variance term and recovers the bulk regime even with positive τ . Our framework assumes the shared- V model, which captures uncertainty about an aggregate population mood rather than uncertainty about individuals; this is the right object to be uncertain about for the puzzle as posed, but the modeling choice is consequential and should be made explicitly. Independence of votes given V is also assumed throughout; weak positive correlation among voters can substantially change the variance structure. The closed-form (14) assumes a Gaussian prior; non-Gaussian priors give the same qualitative phase structure but different constants, with the result depending on the prior density and tail mass at the knife edge. Finally, the analysis is conducted within causal decision theory; alternatives produce different recommendations.

Acknowledgment

This analysis was prompted by a conversation with my close friend David Jin, whom I met as a fellow math student at the University of Michigan. The ideas from that conversation, in particular the use of pivotal voter theory to calculate the implied selfishness/altruism indifference threshold, were then worked through in conversation with Claude Opus 4.7, who corrected my frequent calculation mistakes. The framing, modeling decisions, editorial direction, and verification of results are the author's. As we are all aware by now, AI is, of course, infallible, so any errors are surely my own.

References

- Chappell, R. Y. (2026). Buttons, blenders, and coordination. *Good Thoughts (Substack)*, April 2026. <https://www.goodthoughts.blog/p/buttons-blenders-and-coordination>.
- Downs, A. (1957). *An Economic Theory of Democracy*. Harper and Row, New York.
- Edlin, A., Gelman, A., and Kaplan, N. (2007). Voting as a rational choice: Why and how people vote to improve the well-being of others. *Rationality and Society*, 19(3):293–314.
- Gelman, A., Silver, N., and Edlin, A. (2009). What is the probability your vote will make a difference? *NBER Working Paper No. 15220*.
- Graber, J. (2026). Blue button or red button: some mathematical musings. *God and Game Theory (Substack)*, April 30, 2026. <https://godandgametheory.substack.com/p/blue-button-or-red-button>.
- Kruel, A. (2026). The math and assumptions behind the red-blue thought experiment. *Axis of Ordinary (Substack)*, April 27, 2026. <https://axisofordinary.substack.com/p/the-math-and-assumptions-behind-the>.
- Mulligan, C. B. and Hunter, C. G. (2003). The empirical frequency of a pivotal vote. *Public Choice*, 116(1/2):31–54.
- Myerson, R. B. (1998). Population uncertainty and Poisson games. *International Journal of Game Theory*, 27(3):375–392.
- Riker, W. H. and Ordeshook, P. C. (1968). A theory of the calculus of voting. *American Political Science Review*, 62(1):25–42.
- Urban, T. (2026). Red or blue button thought experiment poll. Post on X, April 24, 2026. <https://x.com/waitbutwhy/status/2047710215265730755>.